2026 NCHC Open Hackathon

NCHC X OpenACC X NVIDIA

Jay Chen
NVIDIA Solution Architect | NCHC Open Hackathon Organizer
jaych@nvidia.com

OpenACC
More Science, Less Programming

OPENACC – CELEBRATING 15 YEARS

Building Community.







Ecosystem Development	Training/Education	OpenACC Specification

OpenACC More Science, Less Programming
OPEN HACKATHONS
NCHC 國家高速網路與計算中心 National Center for High-performance Computing
NVIDIA

NCHC Open Hackathon

Since 2020







2020

2021

2022

2023

2024

2025







OpenACC More Science, Less Programming

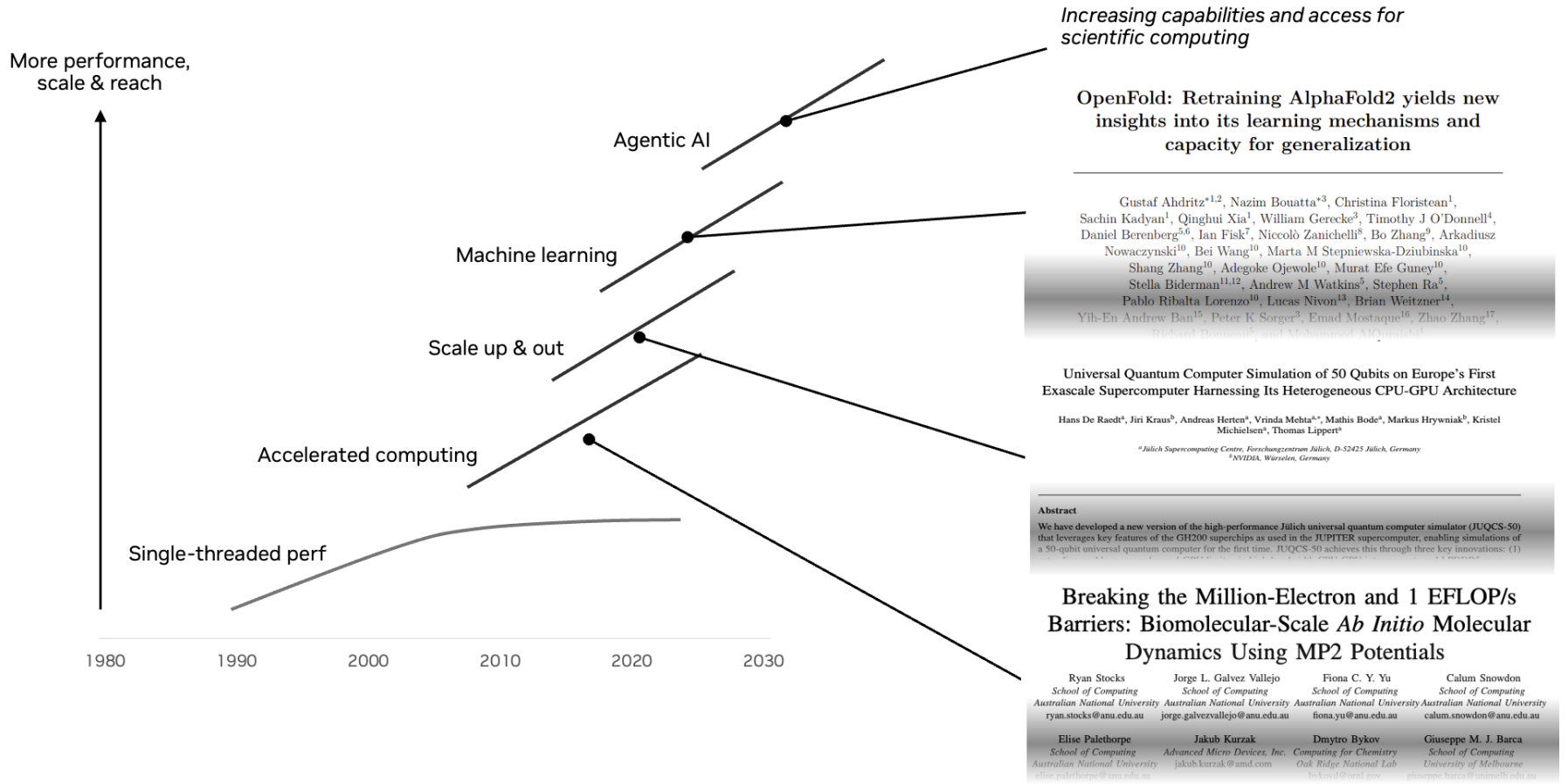
OPEN HACKATHONS

國家高速網路與計算中心 National Center for High-performance Computing

NVIDIA

Incredible Growth in Scale and Reach of Scientific Computing

Expanding capabilities, accessibility and applications leading to new discoveries



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Innovation & R&D Breakthroughs



Talents + High-Potential Projects

WORKFLOW SPEED UP
Agentic Coding

OpenCode +
NCHC RAP LLM API

Bring Your Own Agents

WORKLOAD SPEED UP
NVIDIA Accelerated Computing

Skills + Software

CUDA-X

NVIDIA CPU • GPU • DPU

OPEN HACKATHON PLATFORM

SHARED SUPPORT
MENTORS • HPC COMPUTING RESOURCES

NVIDIA

國家實驗研究院
國家高速網路與計算中心
National Center for High-performance Computing

OPEN HACKATHONS

NCHC × NVIDIA · OPEN HACKATHON

七年協作，讓科研創新加速成真。

2020—2026 · 七年持續推進

國網超級電腦 × NVIDIA CUDA-X × 專家陪跑

2026 再加速：GPU 工作負載 × Agentic AI 工作流程

7 連續年度 2020—2026

73 跨域科研專案 含本屆進行中專案

6 核心應用領域 從科學模擬到 AI

約365 參賽人次 每隊 5 人估算

73+ NVIDIA Mentor 投入 輔導人次・估算

16 今年進行中 Workload × Workflow

地球・流體・天體・電漿 — 14 專案

IES-Geodynamics	2020	Poly-Phys	2020
46×	地球動力學 OpenACC；年度彙整	>10×	物理模擬 CUDA；年度彙整
UniTensor	2020	IES-Geodynamics	2021
3×	張量運算 cuTENSOR；年度彙整	13.6×	地球動力學 OpenACC；年度彙整
TIFDM	2021	THINKLAB	2021
≈2×	波動模擬 OpenACC；年度彙整	672×	地震模擬 Python CUDA 核心
CARL	2021	COSEnu	2021
≈4.5×	流體動力學 OpenACC；年度彙整	40×	物理／微中子 OpenACC；年度彙整
CEAG	2022	COSEnu	2022
8×	爆炸天體物理 OpenACC；多 GPU	6×	微中子傳輸 OpenACC；多 GPU
NTUST CFD Lab	2023	PTSG Taiwan	2025
16.7×	三維 CFD 求解器 主迭代運算	15.1×	低溫電漿 PIC/MCC 粒子移動核心
Team NTU	2026	西灣海環	2026
進行中	MFC／WRF 天氣與流體模擬 雙 KPI・A1-01	進行中	SCHISM 海流模擬 雙 KPI・A1-03

量子・材料・組合最佳化 — 12 專案

AI-Scheduler	2021	長庚大學 CGU	2022
11.3×	智慧製造排程 OpenACC；年度彙整	≈50,000×	退火／護理排班 CUDA 核心 vs NumPy
QKBA	2022	Schrödinger’s cat	2023
≈34×	量子核心分類 cuStateVec；年度彙整	126×	QUBO 求解 端到端／指定測試
haofan2023	2023	Parallel Minds	2024
468×	量子電路模擬 vs 單 CPU 核心	9×	螢火蟲搜尋演算法 指定規模求解測試
NoLab	2024	CYCU_Quantum	2024
8,282×	量子化學 VQE 估算：H12／24 qubits	10,000×	量子 PageRank 指定經典模擬案例
Qa-MolGen	2025	CYCU Power Lab	2025
5,417.9×	量子分子生成 指定電路取樣	109.2×	量子 ML 風場預測 CPU → GPU 訓練
The Quantum Annealers	2026	open_qblock	2026
進行中	GPU 退火／QUBO 求解 雙 KPI・A1-02	進行中	量子位元校正 Agent 雙 KPI・A1-04

氣候・天氣・海洋 — 10 專案

LCCR	2020	AS-LCDM	2020
60×	大氣模擬 OpenACC；年度彙整	1.6×	大氣模擬 OpenACC / CUDA
Finding NEMO 2020	2020	CWA mesh generation	2023
未公布	海洋模式 未列倍速	12×	MPAS 網格生成 領域流程
CWA_GVER	2023	氣象署 × 興大應數聯隊	2023
44×	全球系集模式驗證 指標運算，不含 I/O	1.8×	CWAGFS-TCo 平流 不含 I/O／初始化
氣象署 × 興大應數聯隊	2024	GBA-VVM	2024
70.6×	天氣物理參數化 核心彙整，不含傳輸	18×	大氣平流 平流子程式
Usagi	2025	AI-Taiwan VVM	2025
777×	氣候模擬器矩陣分解 Cholesky 分解	12.8×	天氣 AI 降尺度 小資料集訓練測試

視覺・農業・運動・機器人 — 10 專案

CHTperson	2021	HiFace	2021
3.86×	人流分析 DeepStream	2.59×	人臉辨識 DeepStream
IMAC	2022	Elsa Robotics	2024
未公布	人群辨識 TAO；訓練與準確率改善	11×	機器人導航與感知 管線 0.5 → 5.5 fps
Parallel	2025	NDVIgnition	2025
184.2×	擴散式影像修復 CPU → GPU	5.3×	無人機影像重建 重建階段
Paw Patrol	2025	CropSight	2026
8.5×	三維人體姿態重建 跨 GPU 端到端	進行中	全國農地影像分類 雙 KPI・B4-13
HSCC	2026	Parallel Robotics	2026
進行中	長影片事件定位 雙 KPI・B4-15	進行中	Isaac Lab／四足機器人 雙 KPI・B4-16

語言・音訊・智慧基礎設施 — 13 專案

NTHU-LSALAB	2023	YSS	2023
11×	5G 網路卸載 延遲降低；另省 CPU 15%	2.2×	功能加密／隱私 BSGS 解密
NCHC-Speedrunning	2023	NYCU HPC team2	2024
≈3×	TAIDE／LLM 推論；同 A100 比較	41.7×	NVLM 多模態模型 純文字推論測試
NTUT_BirdSong	2024	NTHU_LSALAB	2024
3.6×	鳥鳴基礎模型 整體訓練比較	1.23×	DPU／GPU 封包處理 封包處理測試
Plantmen	2024	NYCU ・ Bocchi the CUDA!	2025
10×	農業 RAG 聊天機器人 報告端到端／併發測試	4.6×	LLM 強化學習 rollout 取樣
TreeLine	2026	NCKU ITLAB	2026
進行中	DPU／DDoS 偵測 雙 KPI・A2-05	進行中	資料中心 DPU 卸載 雙 KPI・A2-06
Lab 117	2026	Talk2ALMA	2026
進行中	RoCEv2／ICRC 卸載 雙 KPI・A2-07	進行中	天文檔案／LLM 檢索 雙 KPI・B4-12
NTU_SPML	2026		
進行中	語音／口語模型訓練 雙 KPI・B4-14		

健康・生醫・生命科學 — 14 專案

SABID	2020	ALICE	2021
8×	冷凍電子顯微鏡 CUDA；年度彙整	148×	生物資訊 CuPy / cuDF
FlyingTurtle	2021	MI&HIM2021	2021
3.4×	腦部影像 CUDA；年度彙整	未公布	疾病預測 僅有準確率，非倍速
GFEverything	2022	CYCU BME	2023
≈8×	腦部腫瘤分割 MONAI；多 GPU	1.5×	AI 耳鏡診斷 模型推論
WTMH	2023	TXM-AI	2023
40×	心律不整篩檢 長段 ECG	≈30×	X 光背景校正 FFT 損失基準測試
Dream Chaser	2024	smile lab	2024
7.8×	蛋白質分子對接 主辦成果彙整	5.5×	病理全玻片前處理 前處理理論推估
BioVibe-Oncology	2026	PairLab	2026
進行中	癌症抗藥性／多體學 雙 KPI・B3-08	進行中	MPN 多模態病理 AI 雙 KPI・B3-09
為甚麼SSIM不能上升	2026	Lab210	2026
進行中	3T → 7T MRI 合成 雙 KPI・B3-10	進行中	心臟 CT 視覺語言模型 雙 KPI・B3-11

KICK-OFF · WELCOME TO THE COMMUNITY

NCHC NVIDIA Open Hackathon 2026

Welcome, everyone. Every team brings a challenge. Together, we make progress possible.

14 OCTOBER — 4 NOVEMBER 2026

Workload speed-up × Workflow speed-up

NVIDIA platform + agentic AI · Hosted by NCHC

4	16	78	20	4	7
Report groups 每週報告組	Selected teams 入選隊伍	Participants 參賽者	Mentors 導師	Hackathon team 活動執行團隊	Sponsors 支持與贊助者
Two final-day tracks · A / B	A1 · A2 · B3 · B4	Unique people across team rosters	14 PICs · 7 floating · 1 dual role	Coordination & delivery	NVIDIA + NCHC

A1 Scientific simulation & quantum

GROUP HOST **Pika Wang** · 4 teams · 21 member seats

A1-01 Team NTU

MENTOR PIC **Rosie Wu** DevTech

- **YuJen LIU / 劉又甄**
- 孫語辰
- 張皓凱
- 鄭又齊
- 廖昀陽
- 劉蕃熙

A1-02 The Quantum Annealers

MENTOR PIC **Jiun-Cheng Jiang** SA

- **MingZhuo TING / 陳明濯**
- KaiMing WU / 吳楷銘
- ChengYing WU / 吳政穎
- YenJui CHANG / 張晏瑞
- HsiSheng GOAN

A1-03 西灣海環

MENTOR PIC **Shijie Wang** DevTech

- **Chihung WANG / 王啓竑**
- Haoyu ZHENG / 鄭皓宇
- ChiaYing TUNG / 東佳穎
- YiHsuan KUAN / 官宜萱

A1-04 open_qblock

MENTOR PIC **Pika Wang** SA

- **TzuSheng CHIEN / 錢祖晟**
- 彭國忠
- 胡凱翔
- 林俊華
- 李建霖
- 徐義竣

A2 Data-center systems

GROUP HOST **Richard Huang** · 3 teams · 16 member seats

A2-05 孟勳老師剛剛說我們可以，邪萬你說呢

MENTOR PIC **Richard Huang** SA

- **WenJu CHIANG / 蔣汶儒**
- 郭彥廷
- 沈昱宏
- 廖洺玄
- 詹詠翔

A2-06 黑化肥揮發會發灰

MENTOR PIC **Richard Huang** SA

- **ChihYuan CHENG / 鄭智元**
- TsungHsien LIN / 林宗賢
- TseChun WANG / 王澤鈞

A2-07 Lab 117

MENTOR PIC **Richard Huang** SA

- **江宸安**
- 洪祥銘
- 張慕天
- CheYu HUANG / 黃哲宇
- 管培勛
- 林瀚璿
- 李知恆
- 陳靖昀

B3 Biomedical & healthcare AI

GROUP HOST **Yang-Hsien Lin** · 4 teams · 16 member seats

B3-08 BioVibe-Oncology_SLYu's Lab

MENTOR PIC **Ying-Ja Chen** SA

- **YiJing HSIAO / 蕭乂菁**
- 邢釩怩
- SungLiang YU / 俞松良
- 小葵蝦

B3-09 PairLab

MENTOR PIC **Andy Liao** SA

- **ChengKuan LIN / 林政寬**
- ChiaJen LIU
- YiYang FENG / 馮宜陽
- YuChee TSENG
- Tsai,Sung-Hao / 蔡松豪
- Tong-Jin Chou / 周彤瑾

B3-10 為甚麼SSIM不能上升

MENTOR PIC **Yang-Hsien Lin** SA

- **TingYu LEE / 李庭羽**
- 吳宇若

B3-11 Lab210

MENTOR PIC **Warren Tseng** SA

- **滕明哲**
- 盧翊禎
- 張仁杰
- 鄭維勛

B4 Perception, language & embodied AI

GROUP HOST **Virginia Chen** · 5 teams · 27 member seats

B4-12 Talk2ALMA

MENTOR PIC **Virginia Chen** SA

- **TienHao HSIEH / 謝天晧**
- ShihPing LAI
- ChinFei LEE
- ZihChing CHEN
- KuoSong WANG

B4-13 CropSight

MENTOR PIC **Cliff Chiu** SA

- **ShengChieh LAI / 賴聖傑**
- 陳子員
- 劉青慶
- 鄧杰修
- 江承恩
- 晏文芳

B4-14 NTU_SPML

MENTOR PIC **Iven Fu** SA

- **LiangHsuan TSENG / 曾亮軒**
- 陳竣瑋
- Hung-yi Lee
- EnPei HU / 胡恩沛
- Wei-Chih Chen

B4-15 HSCC

MENTOR PIC **Claire Chang** DevTech

- **YuChee TSENG / 曾煜棋**
- PinHsuan CHOU
- 莊昕珉
- 蘇靖淵
- ChengKuan LIN / 林政寬
- YuTing TSENG / 曾友亭
- 謝欣哲

B4-16 Parallel Robotics

MENTOR PIC **Johnson Sun** SA

- **YuChung CHEN / 陳昱仲**
- YuHsiang CHEN / 陳郁翔
- YungShun CHAN / 詹詠順
- MinChih SHIH / 施閔智

Floating mentors

Cross-team expertise · 跨隊協助

- **I-Shu Wang** SA — Bioinformatics and Genomics
- **Chi-Cheng Fu** SA — Healthcare AI
- **Jay Chen** SA — Co-organizer
- **Ken Liao** SA — Medical Imaging
- **Johnson Sun** SA — Agentic Coding
- **Kevin Chen** SA — DPU and DOCA
- **Peggy Chen** SA — CUDA-Q & Robotics

Hackathon team

- **Ganesh Sai Apuroop Koppaka**
- **Apoorva Laharia**
- **Jinny Lin**
- **Bharatkumar Sharma**

Sponsors

Thank you for making it possible

- **Eddie Huang** SA
- **Ping-Sung Ai** SA
- **Vincent Chiu** AM
- **Bai-Cheng Jeng** DevTech
- **Eric Kang** SA
- **Z Wu / 吳宙錦** — NCHC · Joint Lab
- **Kevin Lin** — NCHC · RAP Platform CTO

Many disciplines. One community. Two acceleration goals.

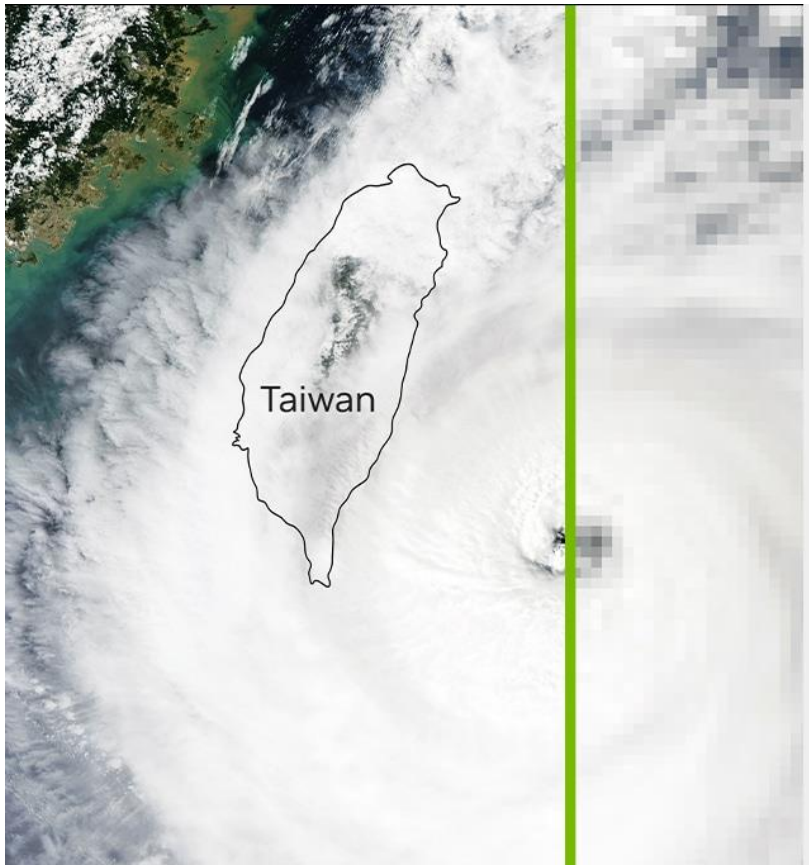
Bold name = primary contact · 粗體為隊伍窗口

Roster: 2026-11-03 · v101 · 80 team-member seats / 78 unique participants. Mentors are deduplicated across PIC and floating roles; role counts may overlap.

NCHC 超級電腦為台灣研究帶來大躍進

220*H200-NVL8 + 2*GB200-NVL72

https://blogs.nvidia.com.tw/blog/taiwan-research-supercomputer/







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OPEN HACKATHONS

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NVIDIA

Invited Opening Speaker



Deputy Director General- Chin-Min Yao

Education：

Ph.D in Mechanical Engineering, National Taiwan University

Experiences：

- Senior Researcher in Director-General's Office, NCHC
- Division Director, Planning and Promotion Division, NCHC
- Adjunct associate professor, Institute of NanoEngineering and MicroSystems, NTHU
- Deputy Director General, NCHC
- Division Director, Computing Application Division, NCHC

Invited Opening Speaker



Eric Kang

NVIDIA Senior Solution Architect Manager

Eric Kang leading the Solutions Architect team in NVIDIA, working with customers and partners to deliver the world's best solutions for artificial intelligence, deep learning and machine learning, professional visualization, and high-performance computing. Specializing in NVIDIA's advanced solutions architecture planning, he has in-depth research on high-performance computing and deep learning computing architecture and participates in relevant important projects architecture planning. Prior to joining NVIDIA, he served in HP Enterprise and as solutions consultant lead for data center transformation practice. Experience including high-performance computing, advanced data center planning, private cloud and public cloud management solutions, data center management automation.

NCHC x NVIDIA Workshops/Bootcamps/Hackathons

Since 2020

Quarterly Hands-on Events

Across ALL Scientific Domains

Including Quantum, Weather, Life Science, CFD, LLM, HPC …etc.



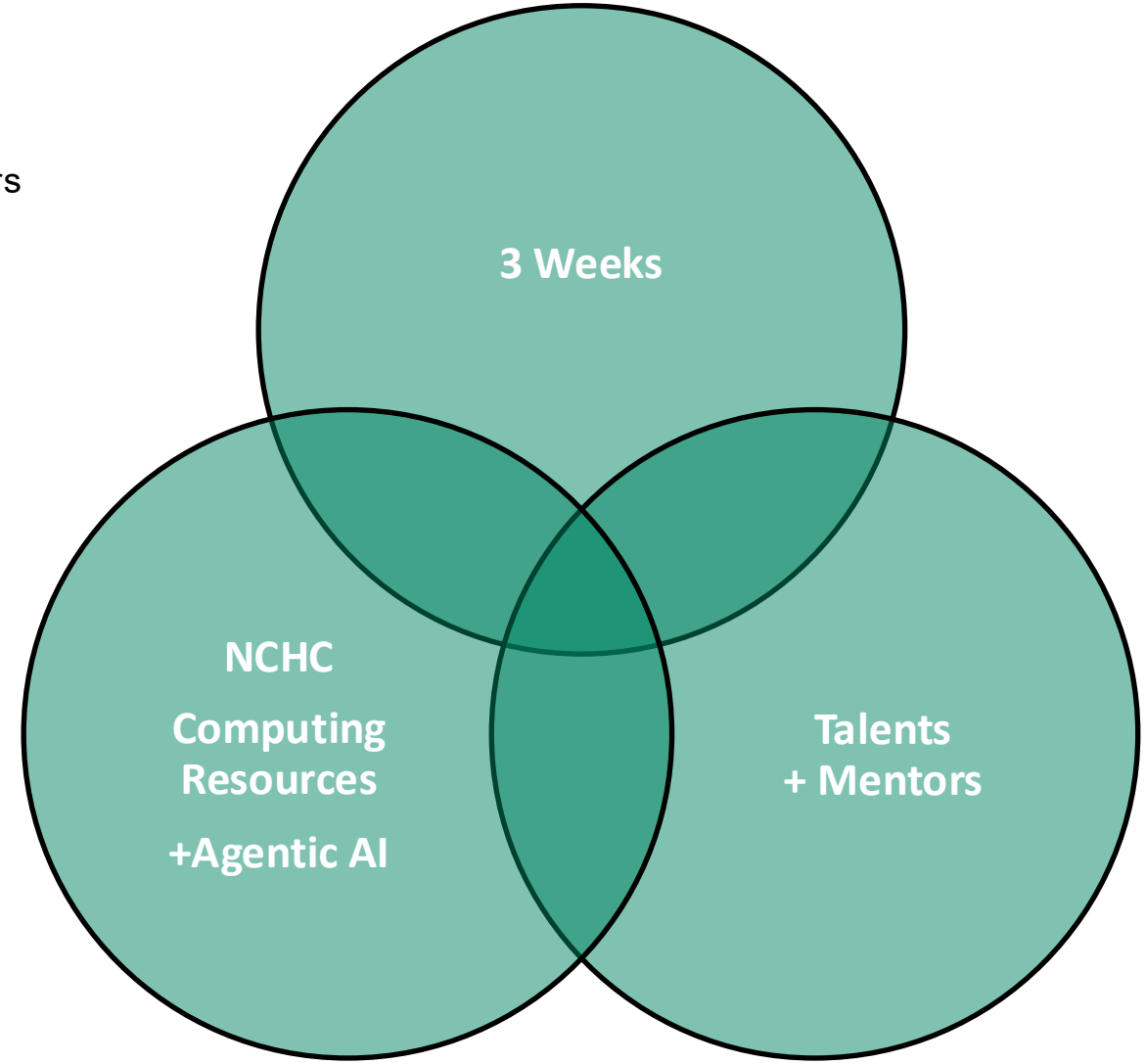
2025 NCHC Open Hackathon

Important Dates

- ~~10 / 22 (Wed)：Day 0: Kick-off Meeting 15:00~17:00 PM (Online)~~
  - ~~5-minute presentation per team~~
- ~~10 / 29 (Wed)：Day 1: Scrum #1 Meeting 15:00~16:00 PM (Online)~~
  - ~~5-minute presentation per team~~
- ~~11 / 05 (Wed)：Day 2: Scrum #2 Meeting 15:00~16:00 PM (Online)~~
  - ~~5-minute presentation per team~~
- **11 / 12 (Wed)：Final Day 10:00~16:00 PM (In-person at NCHC)**
  - **12-minute presentation + 3-minute Q&A per team**

It is not easy

Thanks Teams and Mentors



NCHC NVIDIA OPEN HACKATHON 2026

FINAL PRESENTATION AGENDA

Wednesday, November 4, 2026 · Asia/Taipei

09:30–10:00 | **Opening & Meet up**

10:00–12:30 | **Final Presentation · Parallel Tracks A & B** | 15 minutes per team · 12-min presentation + 3-min Q&A

TRACK A

Host: Jay Chen

10:00–10:15	A1-01	Team NTU
10:15–10:30	A1-02	The Quantum Annealers
10:30–10:45	A1-03	Baytech · SCHISM
10:45–11:00	A1-04	open_qblock
11:00–11:15	A2-06	DPU KV-cache Prefetch
11:15–11:30	A2-07	Lab 117
11:30–11:45	A2-05	TreeLine
11:45–12:30		Discussion / buffer

TRACK B

Host: Yang-Hsien Lin

10:00–10:15	B3-08	BioVibe-Oncology
10:15–10:30	B3-09	Pair Lab
10:30–10:45	B3-10	3T-to-7T MRI
10:45–11:00	B3-11	Lab210
11:00–11:15	B4-12	Talk2ALMA
11:15–11:30	B4-13	CropSight
11:30–11:45	B4-14	NTU_SPML
11:45–12:00	B4-15	HSCC
12:00–12:15	B4-16	Parallel Robotics
12:15–12:30		Discussion / buffer

12:30–13:00 | **Wrap-up & Closing**

13:00–14:00 | **Lunch together ~**

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Presentation & Small Gifts

Vote for the best presentation: https://forms.cloud.microsoft/r/acGMqctFHx

Small gifts will be awarded to the highest-voted team in each track during the final presentations:

- 4 x NVIDIA logo USB-C Cable
- 4 x NVIDIA logo Laptop Tote
- 1 x NVIDIA logo Playing Cards and Pins











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Certification

https://github.com/nqobu/nvidia/blob/main/20231207/Certificate_Of_Attendance.pdf



Publications

https://www.nchc.org.tw/Message?itemid=1008&mid=165
https://www.openacc.org/blog/nchc-employs-open-hackathons-and-bootcamps-usher-





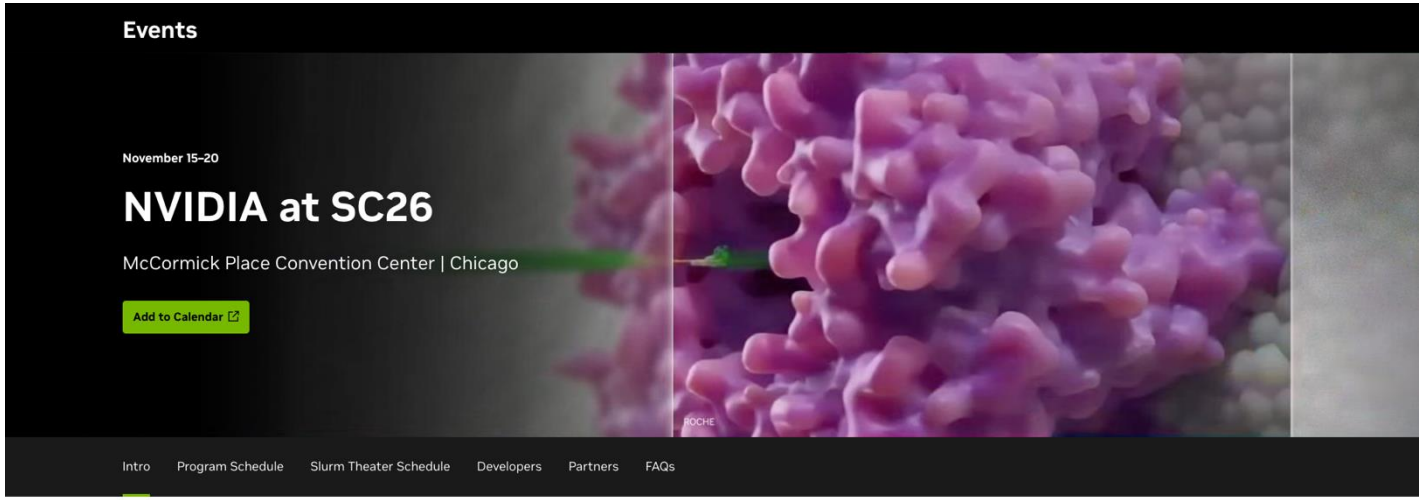
Thanks for Joining 2026 NCHC Open Hackathon

Please send a private message “領取活動獎勵” to @OpenHackBot to get an NVIDIA-sponsored $200 Brev coupon.



SC 2026

https://www.nvidia.com/en-us/events/supercomputing/



Join us at Supercomputing 2026 (SC26) to see how AI-driven scientific workflows, quantum-enabled possibilities, and high-performance platforms are powering tomorrow's complex challenges.

Hear from industry leaders in theater sessions hosted by Slurm in booth #1711 or visit the NVIDIA main booth #3011 to see cutting-edge demos showcasing our latest technologies, engage in activations for developers, and more.



SC26 NVIDIA Build-a-Claw and Ian Buck Special Address

Monday, Nov. 16, 1–5 p.m. CT

Join us for a Build-a-Claw—featuring hands-on demos, connect with the experts, and more.

You can then stay for the NVIDIA SC26 Special Address, where NVIDIA's Ian Buck will highlight the latest cutting-edge breakthroughs revolutionizing high-performance computing.

Schedule:

- 1–3:30 p.m. — Build-a-Claw
- 4–5 p.m. — Special Address

Register Now

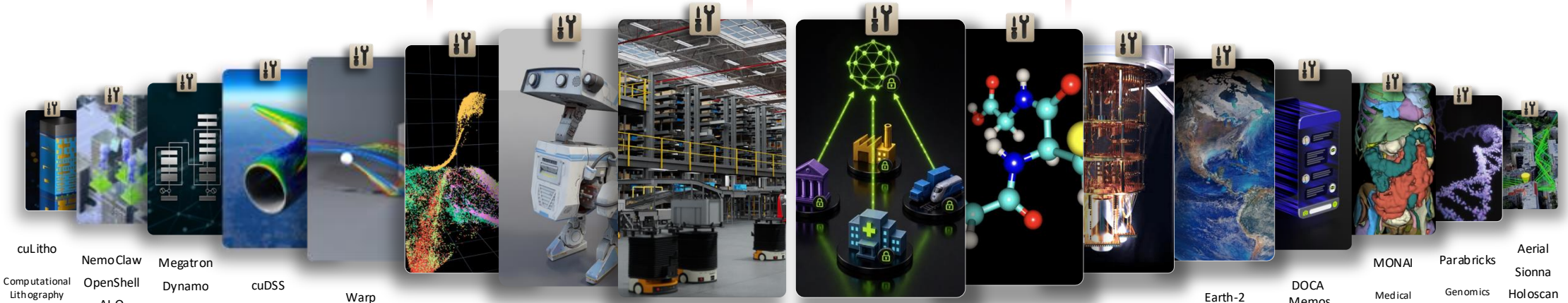
20 Years of CUDA: CUDA-X Agentic Skills for Science

7M
Developers

600M
Downloads

6K
Accelerated Applications

1K
SDKs and Models



cuLitho
Computational Lithography

NemoClaw
OpenShell
AI-Q
VSS
NeMo
Agentic AI

Megatron
Dynamo
cuDNN
CUTLASS
Voice Agent
Deep Learning and Inference

cuDSS
cuSPARSE
cuFFT
AmgX
Computer-Aided Engineering

Warp
PhysicsNeMo
AI Physics

cuOpt
cuDF
cuML
cuVS
Retriever
cuPhoton
Data Science and Processing

Newton
IsaacLab
Robotics

Omniverse
Real-Time Digital Twins

NVFlare
Federated Learning

cuEST
cuEquivariance
cuTensor
ALCHEMI
Quantum Chemistry

cuQuantum
CUDA-Q
Quantum Computing

Earth-2
Weather Analytics

DOCA Memos
Context Memory Storage

MONAI
Medical Imaging

Parabricks
Genomics

Aerial
Sionna
Holoscan
DAQIRI
Signal Processing

Every Industry | One Architecture | CUDA Everywhere

OpenACC More Science, Less Programming
OPEN HACKATHONS
NCHC 國家實驗研究院 國家高速網路與計算中心 National Center for High-performance Computing
NVIDIA

Kind Reminders

- Your iService account will be removed from Event Program **BY FRIDAY**.
- **Please BACKUP YOUR DATA**!
- Send publication materials to Jay!
- Let’s go lunch!